The Factory Floor

AP Human Geography · Unit 7 · No stimulus (like Question 1) · 7 points · Teacher key · GeoArcade FRQ Tutor

No stimulus. Every factory sits where it does for a reason: close to its raw materials, its customers, or cheap workers. When those costs change, whole industries move, and the places they leave behind have to find a new way to make a living.

A. Identify ONE of the three location factors in Alfred Weber's least-cost theory of industrial location. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Transportation costs
- Labor costs
- Agglomeration

MODEL ANSWER

Transportation costs: the cost of moving raw materials to the factory and finished products to the market.

Common mistake: "Good weather and cheap land." Those can matter to real companies, but they aren't Weber's factors. Name one of his three: transportation, labor, or agglomeration.

Why the model works: It names one of Weber's three factors exactly and adds a few words showing what it means.

B. Describe why a bulk-reducing (weight-losing) industry, such as copper smelting, usually locates near its source of raw materials. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Says the finished product weighs less than the raw materials, because waste is removed
- Describes the payoff: processing near the mine cuts transportation (shipping) costs

MODEL ANSWER

T1- A bulk-reducing industry locates near its raw materials because its finished product weighs much less than its inputs.
E2- Copper ore is usually about 1 percent copper or less, so mills and smelters near mines in places like Arizona and Chile remove tons of waste rock to get a little metal.
A3- Processing near the mine means only the lighter copper gets shipped, not heavy waste rock, which keeps transportation costs low.

Common mistake: "Copper smelters are built near copper mines because that is where the copper is." That's circular. Describe WHY: the ore loses most of its weight in processing, so shipping it unprocessed wastes money.

Why the model works: It names the idea (the product weighs far less than the ore), gives proof (about 1 percent copper), and describes the payoff (cheaper shipping).

C. Explain ONE reason the Industrial Revolution began in Great Britain in the 1700s. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: coal, iron, capital, colonies, rivers/canals, workers, inventions, or political stability
- Explains how it helped industry start: powered machines, paid for factories, or moved goods cheaply

MODEL ANSWER

T1- One reason the Industrial Revolution began in Great Britain is that it had large deposits of coal and iron ore.

E2- Coal fueled the steam engines James Watt improved in the 1760s and 1770s, iron built the machines, and canals like the Bridgewater Canal (1761) moved coal cheaply to Manchester.

A3- Because fuel and metal were cheap and close by, British owners could build steam-powered factories before any other country.

Common mistake: "The Industrial Revolution started in Britain because Britain was rich and powerful." Too vague. Name the specific advantage (coal and iron, colonial trade, canals) and explain how it helped factories start.

Why the model works: It names an advantage (coal and iron), gives proof (Watt's steam engines, the Bridgewater Canal), and explains how it let factories start there first.

D. Explain ONE cause of deindustrialization in the Rust Belt of the United States since the 1970s. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a cause: automation, offshoring to lower-wage countries, or foreign competition
- Explains how it cut Rust Belt jobs: lower costs elsewhere, fewer workers needed, plants closed or moved

MODEL ANSWER

T1- One cause of deindustrialization in the Rust Belt is offshoring: companies moved factory work to countries where labor costs much less.

E2- After NAFTA (1994) and China's entry into the World Trade Organization (2001), many makers of auto parts and appliances opened plants in Mexico and China, where wages were far lower than in Ohio or Michigan.

A3- Because the same products could be made more cheaply abroad, companies closed or shrank Rust Belt plants, which led to the loss of hundreds of thousands of factory jobs.

Common mistake: "The factories got old and rusty." That describes the Rust Belt, not its cause. Name a real cause (automation, offshoring, or foreign competition) and explain how it cost jobs.

Why the model works: It names a cause (offshoring), gives specific proof (NAFTA, China joining the WTO), and explains how that cost Rust Belt jobs.

E. Using ONE real Rust Belt city as an example, explain ONE economic or social effect of deindustrialization on that city. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names an effect: job loss, population loss, a shrinking tax base, vacant buildings, or a shift to service jobs
- Uses a real Rust Belt city as the example
- Explains how losing factories caused the effect

MODEL ANSWER

T1- One effect of deindustrialization on Detroit, Michigan, was a huge loss of population, which shrank the city's tax base.

E2- As auto plants closed or moved away, workers lost their jobs and left: Detroit fell from about 1.85 million people in 1950 to about 640,000 in 2020, leaving tens of thousands of houses vacant.

A3- Because far fewer people and businesses paid taxes, the city struggled to pay for streetlights and police, which helped push Detroit into bankruptcy in 2013.

Common mistake: “Detroit became a bad place to live.” That's an opinion, not a specific effect. Name a real effect (like population loss) and explain how closing the auto plants caused it.

Why the model works: It names the effect (population loss, a shrinking tax base), uses a real city with real numbers (Detroit), and explains the chain to bankruptcy.

F. Explain how microloans can promote economic development, especially for women in developing countries. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Says what microloans do: small loans that let poor women shut out of banks start small businesses
- Explains the payoff: income for her family, children's schooling and health, or empowerment

MODEL ANSWER

T1- Microloans promote development by lending small amounts to poor people, especially women, who are shut out of regular banks because they have no collateral.

E2- Bangladesh's Grameen Bank lends mostly to women, who use the money to buy a milk cow, a sewing machine, or a mobile phone to rent to neighbors, then repay from their sales.

A3- Because she now earns her own income, a borrower can spend more on her children's food, health, and schooling and gains a bigger voice in family decisions, which reduces poverty and empowers women.

Common mistake: “Microloans give people money.” That's true of every loan. Explain what the money is FOR (starting a small business) and how that leads to development.

Why the model works: It explains what the loan does (money for a small business), gives a real example (Grameen Bank), and explains the payoff (income for her children and a bigger voice).

G. Explain ONE way ecotourism can support sustainable development, and explain ONE drawback of ecotourism. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Explains how ecotourism brings money or jobs from visitors who come to see nature
- Connects it to sustainability: gives people a reason to protect nature for the future
- Explains a drawback: crowds damage nature, travel pollution, or profits leaving the community

MODEL ANSWER

T1- Ecotourism can support sustainable development because it turns protecting nature into a source of jobs and income.
E2- Costa Rica protects about a quarter of its land in parks and reserves, and tourists pay to see its cloud forests, creating jobs for local guides and lodge owners; in Rwanda, visitors pay $1,500 each to see mountain gorillas.
A3- Because living forests and gorillas earn money year after year, locals have a reason to protect them for future generations; however, one drawback is that heavy tourism can damage the places visitors come to see: crowds trample trails, disturb wildlife, and lodges and roads clear forest, so if visitor numbers aren't limited, ecotourism can destroy the nature that earns the money.

Common mistake: “Ecotourism is good because people like to see animals.” Liking animals isn't a development reason. Explain how tourist money makes protecting nature pay, and add a drawback.

Why the model works: It explains how ecotourism earns money (jobs, permits), connects it to sustainability (people protect what keeps paying), and adds a drawback.
